Revisiting MAE Pre-Training for 3D Medical Image Segmentation: Application to AI CUP 2025

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***Abstract*—Self-supervised learning (SSL) based on Masked Autoencoders (MAE) has achieved strong performance on 2D natural images, but its extension to 3D medical image segmentation is non-trivial due to volumetric data structure, severe class imbalance, and expensive annotations. The recent S3D model revisits MAE pre-training for 3D convolutional neural networks (CNNs), proposing sparse convolutions, sparse normalization, and densification modules so that a 3D CNN encoder can survive heavy masking and outperform the strongest nnU-Net baseline on multiple 3D medical datasets. In this project, we study how to transfer the S3D methodology to a practical challenge: the AI CUP 2025 Heart CT Segmentation competition, which requires accurate 3D segmentation of the heart muscle, aortic valve, and highly long-tail calcification targets from cardiac CT volumes. We design a three-stage pipeline consisting of S3D-style self-supervised pre-training, nnU-Net fine-tuning, and task-specific post-processing using a 3D morphological filter. Our final system achieves a best score of 0.79678 (average of Dice and IoU), ranking 56/568 on the private leaderboard and placing in the top 10%. We also observe that directly transferring S3D pre-training from MRI-oriented settings to CT data leads to limited gains and may even degrade performance, highlighting a strong cross-modality domain shift. We discuss the architectural insights from S3D, analyze the impact of pre-training and post-processing, and outline future directions such as scaling the SSL pre-training, optimizing input patch sizes, and improving robustness to contrast variations.**

***Index Terms*—3D medical image segmentation, self-supervised learning, masked autoencoder, nnU-Net, cardiac CT, AI CUP.**

I. Introduction

3D medical image segmentation is a fundamental problem in computer vision for healthcare, enabling applications such as surgical planning, treatment monitoring, and patient-specific simulation. In cardiac imaging, accurate delineation of the myocardium and aortic valve is especially important. For example, hypertrophic cardiomyopathy can be alleviated by surgically removing excess myocardial tissue, and transcatheter aortic valve replacement (TAVR) relies on precise valve localization to safely deploy the prosthetic device.

The AI CUP 2025 Heart CT Segmentation competition provides a realistic benchmark for this scenario. Participants are asked to perform 3D segmentation on cardiac CT volumes with four semantic classes: background, heart muscle, aortic valve, and calcification. The competition score is the average of Dice coefficient and Intersection-over-Union (IoU), encouraging both overlap quality and spatial precision.

However, 3D medical imaging is challenging: volumetric data have huge resolution, subtle visual cues, severe class imbalance, and expensive annotations. Although self-supervised learning (SSL) has been successful on 2D natural images, directly transferring these methods to 3D medical data is not straightforward.

Recently, Masked Autoencoders (MAE) [1] have emerged as a powerful SSL paradigm. The S3D model (Spark3D-style MAE) revisits MAE pre-training for 3D CNNs and shows that, with appropriate architectural modifications, a 3D CNN-based MAE can outperform transformer-based MAE and even the strongest nnU-Net baseline on several 3D medical datasets. In this project, we attempt to apply the S3D methodology to the AI CUP 2025 Heart CT task, following a three-stage pipeline: pre-training, fine-tuning, and post-processing.

A. Contributions

Our main contributions are:

- We summarize the key differences between 2D vision and 3D medical imaging and explain why transformer-based architectures often underperform compared to 3D CNNs in this domain.
- We review the S3D masked autoencoder design—including sparse convolution, sparse normalization, densification modules, and residual nnU-Net encoders—and adapt it conceptually for cardiac CT segmentation.
- We build a three-stage pipeline for the AI CUP 2025 Heart CT competition: S3D-style self-supervised pre-training, nnU-Net fine-tuning, and a 3D post-processing module that explicitly addresses long-tail calcification predictions.
- We empirically evaluate the pipeline, achieving a top-10% leaderboard result, and provide an analysis of why naive S3D pre-training did not translate into performance gains for this CT-based task.

II. Background and Related Work

A. 2D Vision vs. 3D Medical Imaging

Compared to 2D natural images, 3D medical images such as CT or MRI volumes have several distinguishing characteristics:

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- **Volumetric structure:** data are represented as $H \times W \times D$ volumes with strong depth continuity across slices. Resolutions commonly range from 160^3 to 512^3 voxels, leading to millions of points per scan.
- **Subtle visual cues:** medical images are typically grayscale with low contrast and weak textures, making object boundaries less obvious than in natural images.
- **Severe class imbalance:** lesions or calcification regions often occupy $< 1\%$ of the volume, while background and healthy tissues dominate.
- **Expensive annotations:** high-quality voxel-wise labels require expert clinicians and substantial annotation time.

Table I summarizes these differences.

TABLE I
COMPARISON BETWEEN 2D VISION TASKS AND 3D MEDICAL IMAGING.

Aspect	2D Images	3D Medical Imaging
Data structure	$H \times W$	$H \times W \times D$
Depth continuity	None	Strong across slices
Resolution	256–1024 px	160^3–512^3 voxels
Cues	Color, texture	Low contrast, subtle edges
Annotation cost	Low/medium	High (voxel-wise)
Class imbalance	Moderate	Severe (long-tail)

B. Transformers for 3D Segmentation

Vision Transformers (ViTs) and DETR-like architectures have achieved strong results on 2D vision. However, in 3D medical imaging they face several limitations:

- **Quadratic attention complexity:** self-attention scales quadratically with the number of tokens. In 3D, the patch grid is huge, quickly causing GPU memory explosion.
- **Data demands:** ViTs often require large-scale training data to learn inductive biases from scratch, which is difficult in the medical domain.
- **Voxel-level precision:** achieving precise 3D boundaries benefits from strong locality and morphological priors, which are naturally encoded in CNNs but absent in standard transformers.

Empirical studies show that nnU-Net [2] and related 3D CNN architectures still dominate benchmarks for medical segmentation, especially when data are scarce.

C. Masked Autoencoders (MAE)

Masked Autoencoders (MAE) [1] pre-train models by reconstructing heavily masked inputs. The key idea is:

- Randomly mask a high proportion (e.g., 75%) of input patches.
- Encode only the visible patches, forcing the encoder to capture global structures.
- Use a lightweight decoder to reconstruct the full input, including masked patches.

On 2D images, MAE is typically paired with ViTs and leads to strong transfer performance. However, directly applying MAE to 3D CNNs tends to fail because CNNs expect dense inputs and local continuity; heavy random masking breaks convolutional assumptions and often leads to feature collapse.

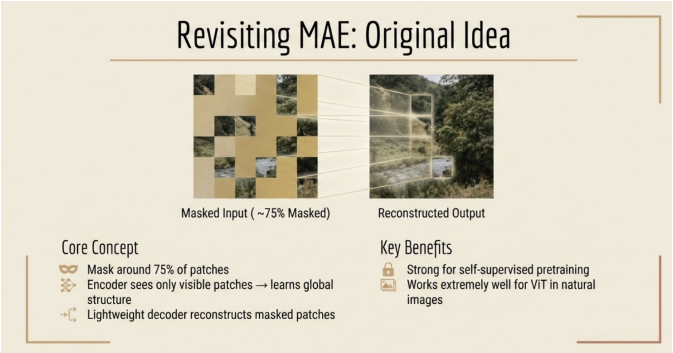

Fig. 1. Conceptual illustration of MAE pre-training: (left) heavily masked 3D input; (right) reconstructed volume.

D. S3D: MAE for 3D CNNs

The S3D model (Spark3D-style MAE) revisits MAE pre-training for 3D medical image segmentation by modifying the CNN architecture to be mask-aware:

- **Sparse convolution:** masked voxels are treated as absent, and 3D convolutions operate only on visible voxels, avoiding feature collapse under high masking ratios.
- **Sparse normalization:** normalization statistics are computed only over visible voxels, preventing unstable activations when most voxels are missing.
- **Mask token densification:** since CNNs cannot naturally insert trainable tokens into the grid, S3D introduces placeholder features for masked voxels and uses a $3\times3\times3$ densification convolution to produce a dense 3D feature map.
- **Residual nnU-Net encoder:** a residual nnU-Net backbone is used as the encoder, and a 75% masking ratio is found to be optimal for downstream performance.

S3D demonstrates that with these modifications, a 3D CNN-based MAE can outperform ViT-based 3D MAE and other SSL baselines, improving Dice scores by approximately 3 points over a strong nnU-Net baseline on several datasets. As shown in Fig. 2, the proposed Spark3D method achieves state-of-the-art results compared to existing approaches.

III. AI CUP 2025 HEART CT SEGMENTATION TASK

A. Problem Definition

The AI CUP 2025 competition focuses on 3D semantic segmentation of cardiac CT scans. Each input is a volumetric CT image of the heart, and the goal is to predict a 3D label map with four classes:

- Class 0: background,
- Class 1: heart muscle,
- Class 2: aortic valve,
- Class 3: calcification.

Heart muscle occupies a large contiguous region, while the aortic valve and calcification are small structures inside or near

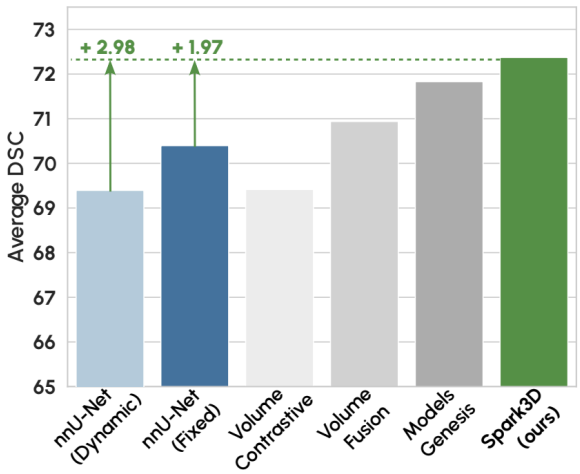

Fig. 2. Well-configured MAE pre-training for CNNs is state-of-the-art. When comparing our MAE pre-trained model on all eleven test Datasets, Spark3D achieves almost +3 DSC points compared to the strong nnU-Net baseline and outperforms current SSL methods.

the heart. Calcifications form a textbook long-tail class: they occupy only a tiny volume fraction but are clinically critical.

B. Evaluation Metric

The competition evaluates models using the average of Dice coefficient and IoU:

$$\text{Score} = \frac{1}{2}(\text{Dice} + \text{IoU}), \tag{1}$$

where both Dice and IoU range from 0 to 1. Higher scores indicate better segmentation quality.

C. Dataset Characteristics and Visualization

To understand both the imaging properties and the intrinsic difficulty of the task, we analyze the AI CUP 2025 cardiac CT dataset using a dedicated visualization pipeline (`visualize_data.py`). We combine representative single-case visualizations with aggregate statistics across the training set to highlight key challenges relevant to model design.

a) Intensity distribution and acquisition characteristics.: Cardiac CT volumes exhibit a wide Hounsfield Unit (HU) range, spanning from -1024 to above 3000 HU across the dataset. Fig. 3 shows the voxel intensity histogram of a representative volume (`patient0001`), where peaks near -1000 HU correspond to air and lung regions, while soft tissues and blood occupy mid-range intensities. The long high-intensity tail reflects dense bone and calcified structures, which are critical for the calcification class. Aggregate statistics in Fig. 4 further reveal substantial inter-patient variability in maximum intensity values, underscoring the importance of robust intensity normalization and partially explaining the difficulty of transferring self-supervised representations trained on other modalities such as MRI.

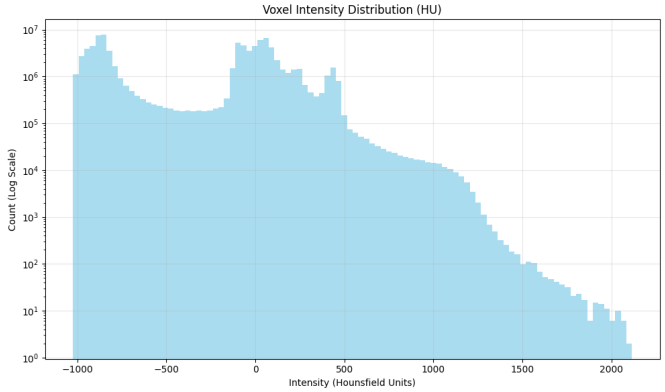

Fig. 3. Voxel intensity distribution (HU) for a representative CT volume (`patient0001`). The y-axis uses a log scale to emphasize both dominant tissues and the high-intensity tail.

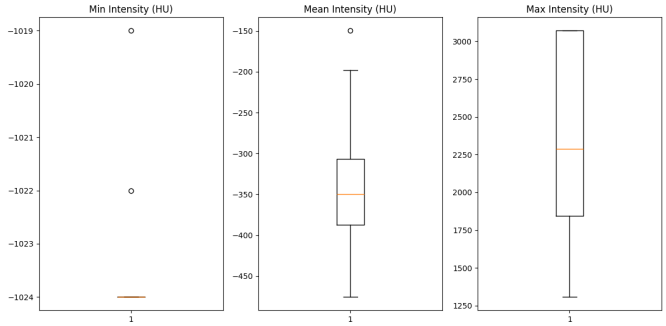

Fig. 4. Aggregate per-volume intensity statistics across the training set, showing the distribution of minimum, mean, and maximum HU values.

b) Spatial resolution and 3D geometry.: The dataset consists of high-resolution volumetric scans with typical shapes around $(512, 512, 337)$ voxels. Voxel spacing is anisotropic: in-plane spacing (x, y) is relatively consistent with a mean of 0.392 mm, while slice spacing (z) is coarser at approximately 0.464 mm (Fig. 5). This anisotropy weakens through-plane continuity and motivates the use of 3D CNNs capable of modeling volumetric context under unequal resolution. Orthographic slices with label overlays (Fig. 6) illustrate the complex, non-convex geometry of the myocardium and the embedded nature of smaller targets. The corresponding 3D surface rendering (Fig. 7) further highlights the intricate spatial relationships among cardiac structures.

c) Severe class imbalance and task difficulty.: Aggregate voxel statistics reveal extreme class imbalance (Fig. 8). Background voxels account for 95.92% of the dataset, while myocardium occupies approximately 4.01%. The aortic valve represents only 0.07%, and calcification is an extreme long-tail class with fewer than 0.01% of voxels. This imbalance makes the task highly sensitive to spurious predictions and motivates both long-tail-aware learning strategies and post-processing to suppress isolated false positives.

d) Summary.: In summary, the dataset is characterized by wide intensity variability, anisotropic high-resolution volumes,

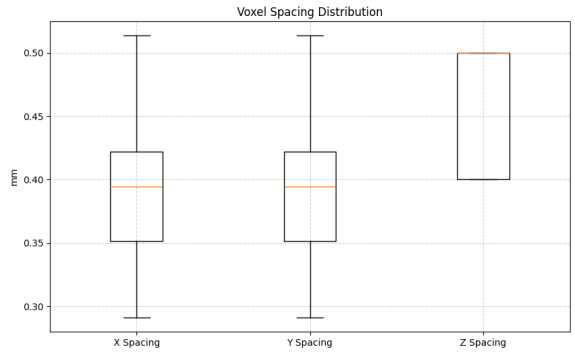

Fig. 5. Voxel spacing distribution along the x, y, and z axes across the training set.

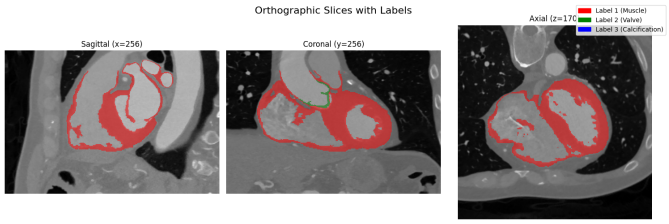

Fig. 6. Orthographic views with segmentation overlays for a representative volume. Red: myocardium, green: aortic valve, blue: calcification.

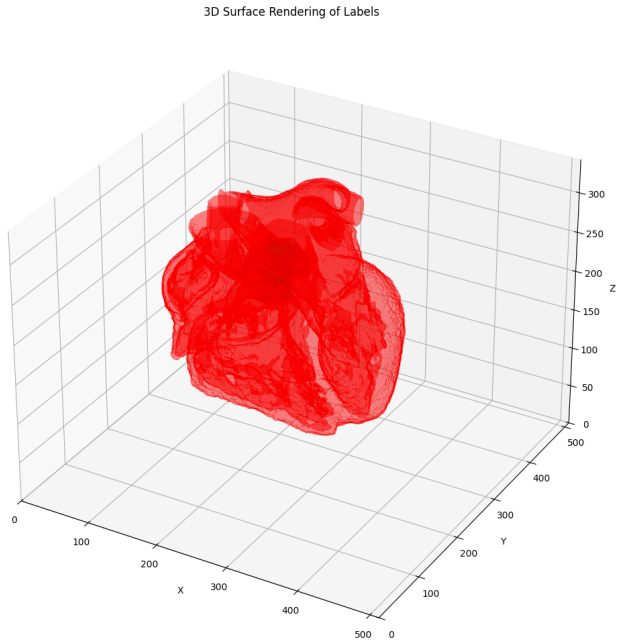

Fig. 7. 3D surface rendering of segmentation labels, illustrating the complex volumetric morphology of cardiac structures.

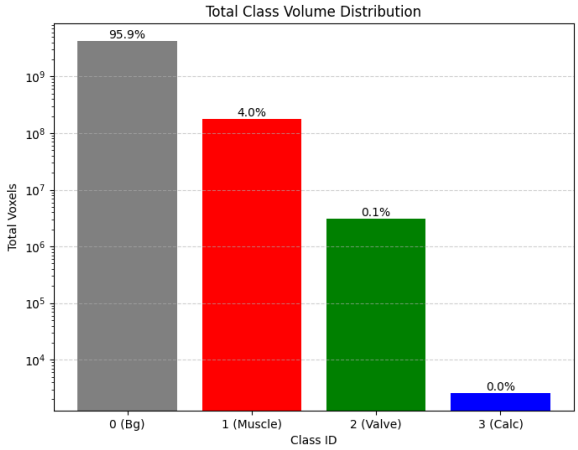

Fig. 8. Aggregate voxel count distribution across classes (log scale). Calcification forms an extreme long-tail class.

and extreme class imbalance. These properties jointly justify the use of 3D CNN-based architectures with strong inductive bias, explain the challenges of cross-modality self-supervised transfer, and motivate our post-processing design for robust segmentation of rare structures.

IV. Methodology

Our approach follows a three-stage pipeline illustrated in Fig. 9, consisting of (1) S3D-style self-supervised pre-training on unlabeled CT volumes, (2) supervised nnU-Net fine-tuning on labeled AI CUP data, and (3) task-specific post-processing to suppress long-tail calcification noise.

A. Overview of the Pipeline

As shown in Fig. 9, Stage 1 learns a strong volumetric representation from unlabeled CT data via masked reconstruction. The pre-trained encoder weights are then transferred to a standard nnU-Net segmentation model in Stage 2. Finally, Stage 3 applies a lightweight 3D morphological post-processing step to improve robustness on extremely sparse calcification regions.

B. Stage 1: S3D-Style Self-Supervised Pre-Training

The goal of Stage 1 is to pre-train a 3D encoder that captures global anatomical structure from CT volumes without relying on manual annotations. Given an unlabeled CT volume, we randomly mask approximately 75% of voxels and train the model to reconstruct the original input from the remaining visible voxels (Fig. 9, top row).

a) Sparse encoder.: The encoder adopts a 3D residual nnU-Net backbone augmented with sparse operations. Specifically, masked voxels are treated as absent rather than zero-valued:

- **SparseConv3D** aggregates features only from visible voxels, preventing feature collapse under high masking ratios.

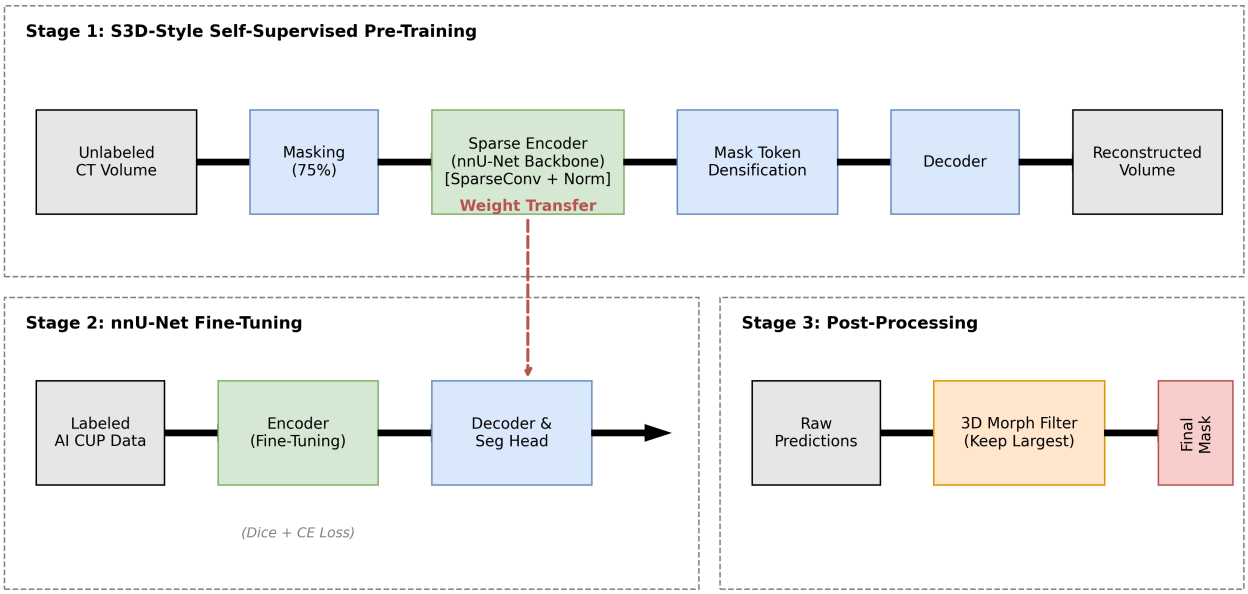

Fig. 9. Overview of the proposed three-stage pipeline: (Stage 1) S3D-style self-supervised pre-training with sparse encoding and masked reconstruction, (Stage 2) nnU-Net fine-tuning for multi-class segmentation, and (Stage 3) 3D post-processing to suppress long-tail calcification noise.

- **SparseNorm** computes normalization statistics using visible voxels only, avoiding unstable activations caused by missing data.

This design allows the encoder block in Fig. 9 to remain effective even when most of the input volume is masked.

b) Mask token densification and decoder.: Since CNNs cannot natively insert learnable mask tokens as in Vision Transformers, we introduce a densification step after sparse encoding. Placeholder features are inserted at masked voxel locations, followed by a $3\times3\times3$ convolution to convert sparse features into a dense 3D feature map. A lightweight decoder with U-Net-style upsampling then reconstructs the original CT volume (Fig. 9, top row). Due to limited computational resources, our implementation follows the nnSSL/S3D reference configuration, which was originally tuned for MRI data; this mismatch later manifests as a domain shift issue in CT fine-tuning.

C. Stage 2: nnU-Net Fine-Tuning

In Stage 2, the pre-trained encoder weights are transferred to a standard 3D nnU-Net segmentation model (Fig. 9, bottom left). The architecture consists of a residual encoder initialized from S3D pre-training and a symmetric decoder with skip connections. The model predicts four classes: background, myocardium, aortic valve, and calcification.

Training is performed on labeled AI CUP Heart CT data using a combination of Dice loss and cross-entropy loss to balance region overlap and voxel-wise classification. Data augmentation, including random cropping, flipping, and intensity perturbations, follows nnU-Net defaults due to limited time for extensive hyperparameter tuning.

D. Stage 3: Post-Processing for Long-Tail Calcification

Calcification is an extreme long-tail class and is particularly sensitive to isolated false positives. To address this, we apply a simple but effective 3D post-processing step to the raw predictions (Fig. 9, right). Specifically, we perform connected component analysis on the predicted label map, apply a $3\times3\times3$ morphological filter to remove isolated voxels and tiny clusters, and discard components below a size threshold. This introduces a spatial prior favoring compact, clinically plausible calcification regions and leads to a consistent improvement in the final evaluation score.

V. Experiments

A. Experimental Setup

Dataset. We use the official AI CUP 2025 Heart CT training set for supervised learning. Unlabeled CT volumes provided by the competition are used for self-supervised pre-training where applicable. Final performance is evaluated by the competition server on hidden test cases using the average of Dice and IoU.

Baseline. We evaluate the following configurations:

- **nnU-Net (baseline):** standard 3D nnU-Net trained from scratch using labeled data only.
- **nnSSL + nnU-Net:** nnU-Net encoder initialized from S3D-style self-supervised pre-training and fine-tuned on labeled data.
- **nnU-Net + Post-processing:** baseline nnU-Net followed by the proposed 3D morphological post-processing.

Implementation Details. All models are implemented in PyTorch, following nnU-Net and nnSSL reference implementations. Experiments are conducted on GPUs with 12 GB

VRAM, which constrains patch size and batch size. Unless otherwise specified, we adopt nnU-Net default training schedules and augmentations.

B. Quantitative Results

Table II reports the competition scores (average of Dice and IoU).

TABLE II
LEADERBOARD SCORES ON AI CUP 2025 HEART CT (AVERAGE OF DICE AND IOU).

Method	Score	Remark
nnU-Net (baseline)	0.79676	From-scratch training
nnSSL + nnU-Net	0.79042	With S3D pre-training
nnU-Net + Post-proc	**0.79678**	Baseline + 3D filter

Our best public score is 0.787232, and private score is 0.79678, achieved by nnU-Net combined with our 3D post-processing. This corresponds to rank 56/568 on the private leaderboard and 36/568 on the public leaderboard, placing us in the top 10% of teams.

Surprisingly, the S3D-style pre-training configuration (nnSSL + nnU-Net) yields a lower score (0.79042), slightly below the from-scratch nnU-Net baseline.

C. Qualitative Results

a) Overall segmentation quality.: Across test cases, the nnU-Net baseline already demonstrates strong performance in delineating the myocardium, producing large, spatially coherent volumes. This confirms that the convolutional inductive bias of 3D CNNs is well-suited for capturing global cardiac anatomy.

b) Multi-view inspection reveals internal false positives.: Fig. 10 visualizes the predicted 3D segmentation results for Patient 52 and Patient 61 using point-cloud projections. For Patient 52, two different viewing angles are shown. From a surface-level perspective, the prediction appears clean and plausible. However, rotating the volume reveals numerous isolated internal prediction points that are not visible from a single viewpoint.

This observation highlights an important challenge: small calcification-related false positives may be hidden inside the myocardium and cannot be reliably detected by inspecting only surface projections or single 2D slices. Effective evaluation and refinement therefore require true 3D spatial analysis.

c) Spatial filtering via post-processing.: To address this issue, we apply connected component analysis followed by a $3 \times 3 \times 3$ morphological kernel. This operation explicitly removes isolated single-voxel predictions and tiny spurious clusters, which are highly likely to be false positives for the extremely sparse calcification class.

As illustrated in Fig. 10, this spatial filtering step effectively cleans the internal prediction space while preserving anatomically plausible structures. The result is not only visually more coherent but also leads to measurable improvements in Dice and IoU.

VI. DISCUSSION

A. Why Did S3D Pre-Training Underperform?

While S3D-style masked autoencoder pre-training has demonstrated strong gains on MRI datasets, its direct transfer to our CT-based cardiac segmentation task did not improve performance. In our experiments, nnSSL + nnU-Net achieved a lower score than the nnU-Net baseline trained from scratch.

We attribute this behavior to several factors.

- **Cross-modality domain shift:** S3D and related self-supervised frameworks are primarily tuned on MRI data, whose intensity distributions, noise characteristics, and contrast mechanisms differ fundamentally from CT. As a result, representations learned under MRI-oriented objectives may not align well with CT-specific semantics without careful adaptation.
- **Limited pre-training scale:** Masked autoencoder-based self-supervised learning typically benefits from very large datasets and long training schedules. Due to limited compute resources and a relatively small pool of unlabeled CT volumes, our pre-training setup may not have been sufficient to learn robust modality-specific representations.
- **Suboptimal hyperparameters for CT:** Hyperparameters such as the 75% masking ratio and patch size were inherited from MRI-centric configurations. Without extensive ablation, these settings may place the model in a suboptimal regime for CT, where fine-grained intensity cues and sharp boundaries are more critical.

Taken together, these observations highlight that self-supervised learning is not guaranteed to improve downstream performance in every setting. Effective SSL requires careful adaptation to the target modality, data scale, and computational constraints, especially in cross-modality scenarios.

B. When Simplicity Wins under Constraints

In contrast to self-supervised pre-training, our task-specific 3D post-processing strategy is simple yet consistently effective. The calcification class exhibits an extreme long-tail distribution, where even a small number of false positives can significantly degrade both Dice and IoU.

By applying connected component analysis followed by a $3\times3\times3$ morphological filter and component size thresholding, we explicitly remove isolated single-voxel predictions and tiny spurious clusters that are spatially implausible. This operation acts as a strong anatomical prior, enforcing spatial consistency at negligible computational cost.

Qualitative 3D visualizations further confirm that many calcification-related false positives are hidden inside the myocardium and are not easily detectable from surface-level inspection or single 2D slices. The proposed post-processing step effectively cleans these internal prediction spaces while preserving clinically plausible structures.

These results suggest that, under tight resource and time constraints, carefully designed domain-aware post-processing can offer a superior cost-performance trade-off compared to complex self-supervised representation learning schemes.

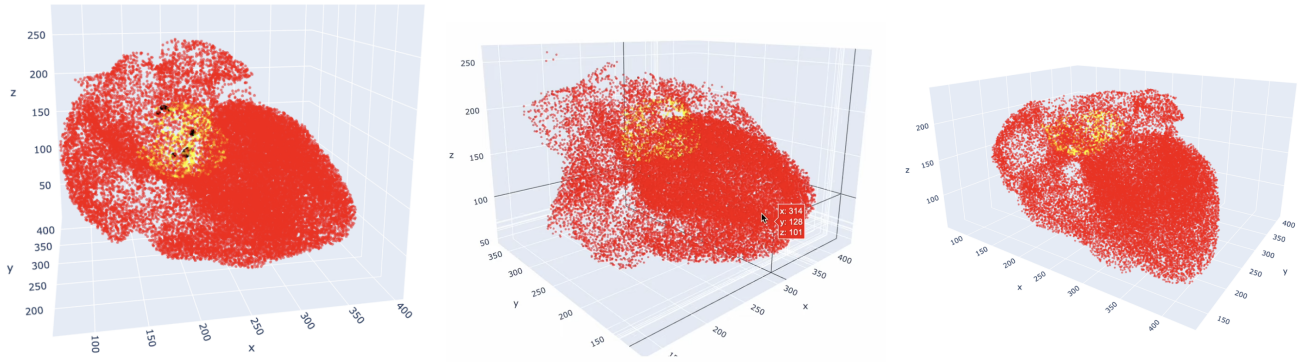

Fig. 10. 3D point-cloud visualizations of predicted segmentations after nnU-Net inference. (**Left, Middle**) Patient 52 shown from two different viewing angles. Small internal calcification-related predictions are only visible from specific orientations, illustrating that surface inspection alone is insufficient. (**Right**) Patient 61, where post-processing removes scattered noise while preserving spatially consistent internal clusters. Red denotes myocardium predictions, while highlighted internal points correspond to aortic valve and calcification candidates.

VII. Conclusion and Future Work

We revisited S3D masked autoencoder pre-training for 3D medical image segmentation and applied it to the AI CUP 2025 Heart CT competition within a three-stage pipeline: S3D-style self-supervised pre-training, nnU-Net fine-tuning, and 3D post-processing. Our final system achieved a best score of 0.79678 and ranked in the top 10% of teams.

However, we also observed that naive S3D pre-training from MRI-oriented settings did not yield performance gains on CT. Our strongest result came from a robust nnU-Net baseline combined with a simple yet effective post-processing module that specifically targets long-tail calcification noise.

A. Future Work

Future directions include:

- **Scaling pre-training:** exploring larger-scale S3D pre-training on cardiac CT, including more unlabeled data and longer training schedules.
- **Patch size and architecture search:** performing systematic ablations over patch sizes, depth, and width of the encoder-decoder, potentially leveraging neural architecture search (NAS).
- **Robustness to intensity and contrast:** evaluating SSL robustness under histogram equalization, different intensity windows, and domain-specific normalizations.
- **Domain adaptation:** applying domain adaptation techniques to bridge MRI and CT representations, such as feature alignment losses, adversarial training, or cross-modality consistency constraints.

We hope that our findings provide a practical reference for future work on self-supervised 3D medical segmentation and competition pipelines.

Acknowledgment

We thank the instructor and teaching assistants of *Computer Vision Practice with Deep Learning* at National Taiwan University for providing the AI CUP 2025 Heart CT dataset, computational resources, and valuable feedback during the project.

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Appendix

This project was completed through close collaboration among all team members. Each member contributed to both the conceptual development and technical execution of the project.All authors contributed to discussions and revisions of the final manuscript. The primary responsibilities are summarized as follows.

- **Jia-Le Liao:** Project ideation, data preprocessing, literature review, and implementation of state-of-the-art (SOTA) models.
- **Yu-Chi Ko:** Project ideation, data visualization, slide preparation, report writing, and comparative analysis of different model configurations.
- **Yao-Kai Li:** Project ideation, accumulation and organization of experimental results, and preparation of presentation slides.
- **Yin-Kuan Li:** Project ideation, literature review, baseline model implementation, and model training and evaluation.